Lecture 20

More Dynamic Programming Examples

Source: Introduction to Algorithms, CLRS

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Example: Let $S =$ "abrakadabra" be a sequence.

Some subsequence of S are : "arkdra", "a", "aaaaa", "abrakadabra".

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Some common sequences of X and Y are: "iit", "dr", "tor".

Some longest common sequences of X and Y are: "iitor", "iitdr".

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- LCS of two sequences or strings is a measure of how similar they are.
- Used to find similarity of DNAs which can be seen as strings of “A”, “C”, “G”, and “T” characters which represent nucleotides.

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Generating all the subsequences takes $O(2^{|x|})$ time.

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Is $LCS(i, j) = LCS(i-1, j-1) + 1$?

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Is $LCS(i, j) = max(LCS(i-1, j), LCS(i, j-1))$?

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Proof: Suppose length of Z is k and, hence, length of $\mathbf{LCS}(x_1 x_2 \ldots x_{m-1}, y_1 y_2 \ldots y_{n-1})$ is $k - 1$.

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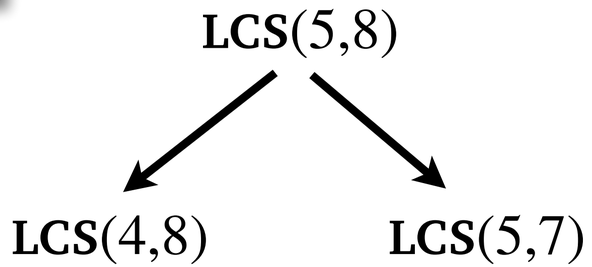

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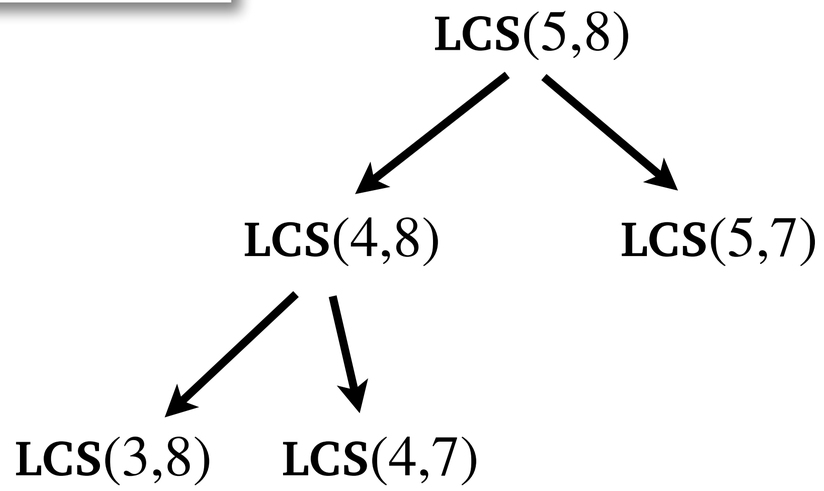

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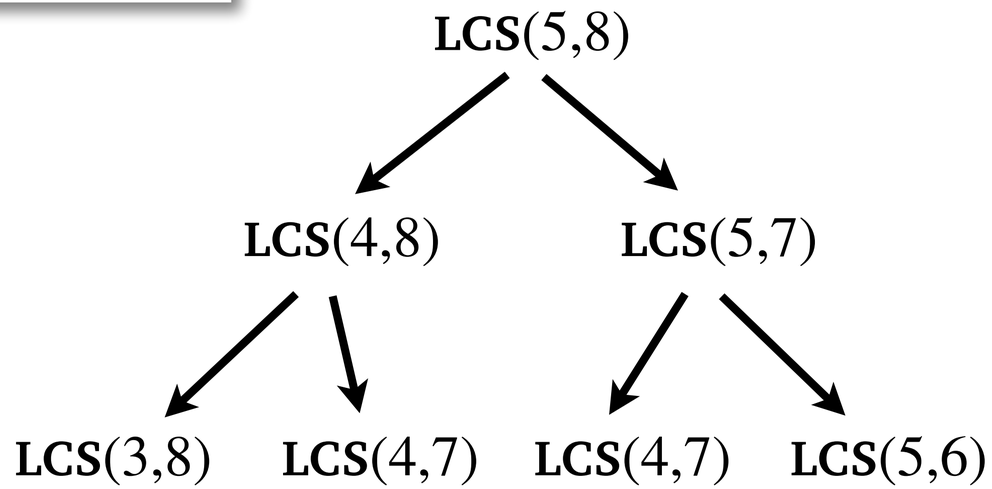

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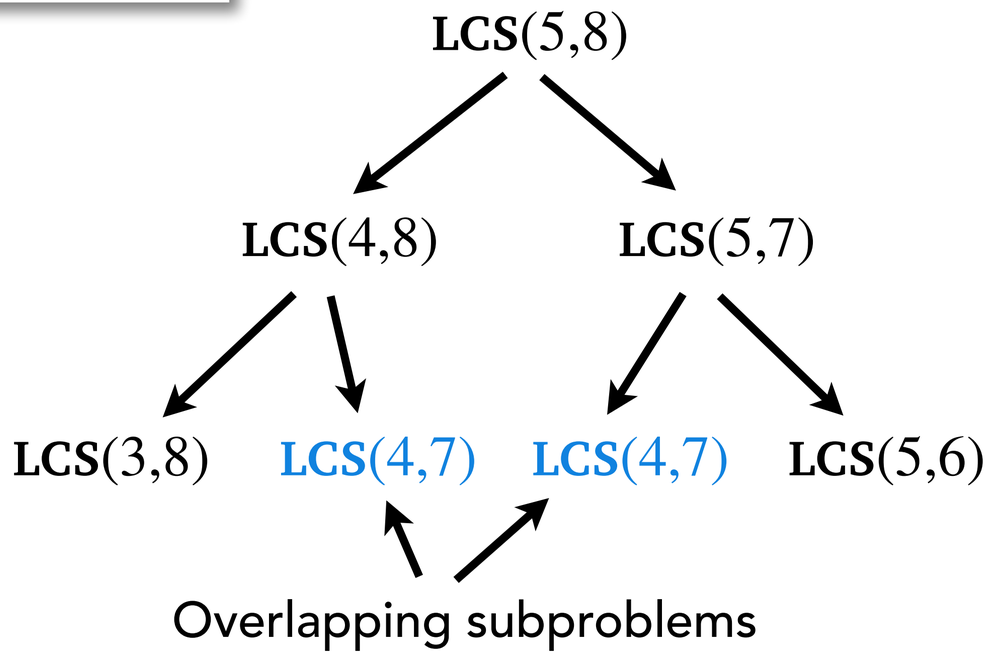

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Time Complexity: $O(nm)$

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dp

0	0	0	0	0	0	0
0						
0						
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0						
0						

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0	0	0	0	0	0	0
0						
0						
0						
0						
0						

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dp

0	0	0	0	0	0	0
0						
0						
0						
0						
0						

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0	0	0	0	0	0	0
0						
0						
0						
0						
0						

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dp

0	0	0	0	0	0	0
0						
0						
0						
0						
0						

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0						
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0	0	0	0	0	0	0
0						
0						
0						
0						
0						

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0	0	0	0	0	0	0
0						
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0						
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0	0	0	0	0	0	0
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0	0	0	0	0	0	0
0						
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0	0	0	0	0	0	0
0						
0						
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0						
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0	0	0	0	0	0	0
0						
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0	0	0	0	0	0	0
0						
0						
0						
0						
0						

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0	0	0	0	0	0	0
0						
0						
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0						

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0	0	0	0	0	0	0
0						
0						
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0						
0						

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0	0	0	0	0	0	0
0						
0						
0						
0						
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0						
0						
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0	0	0	0	0	0	0
0						
0						
0				(i, j)		
0						
0						

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0	0	0	0	0	0	0
0						
0						
0				(i, j)		
0						
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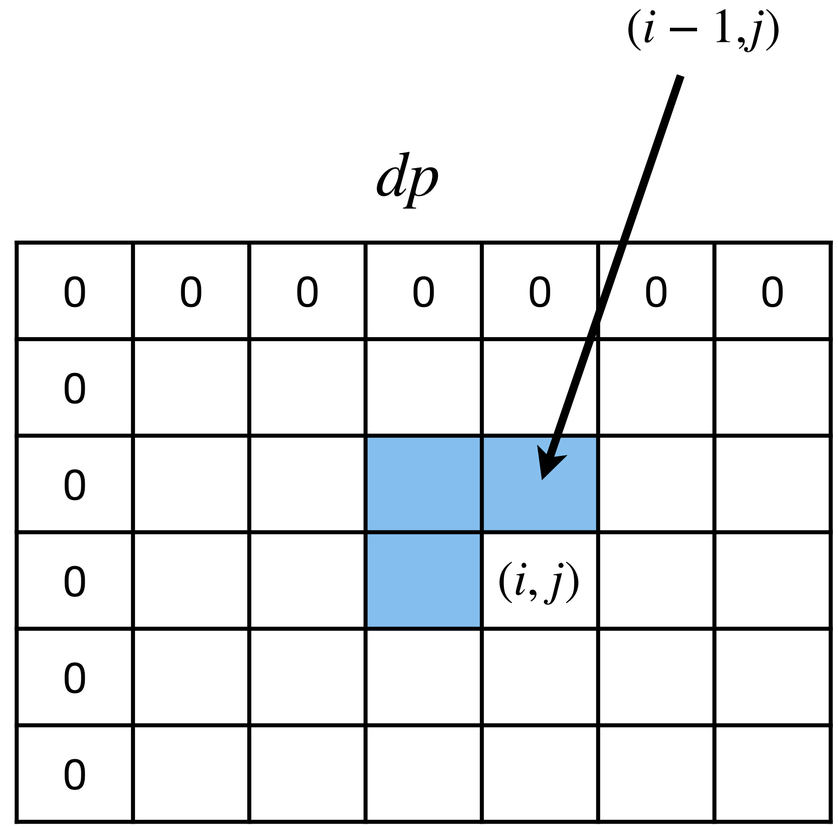

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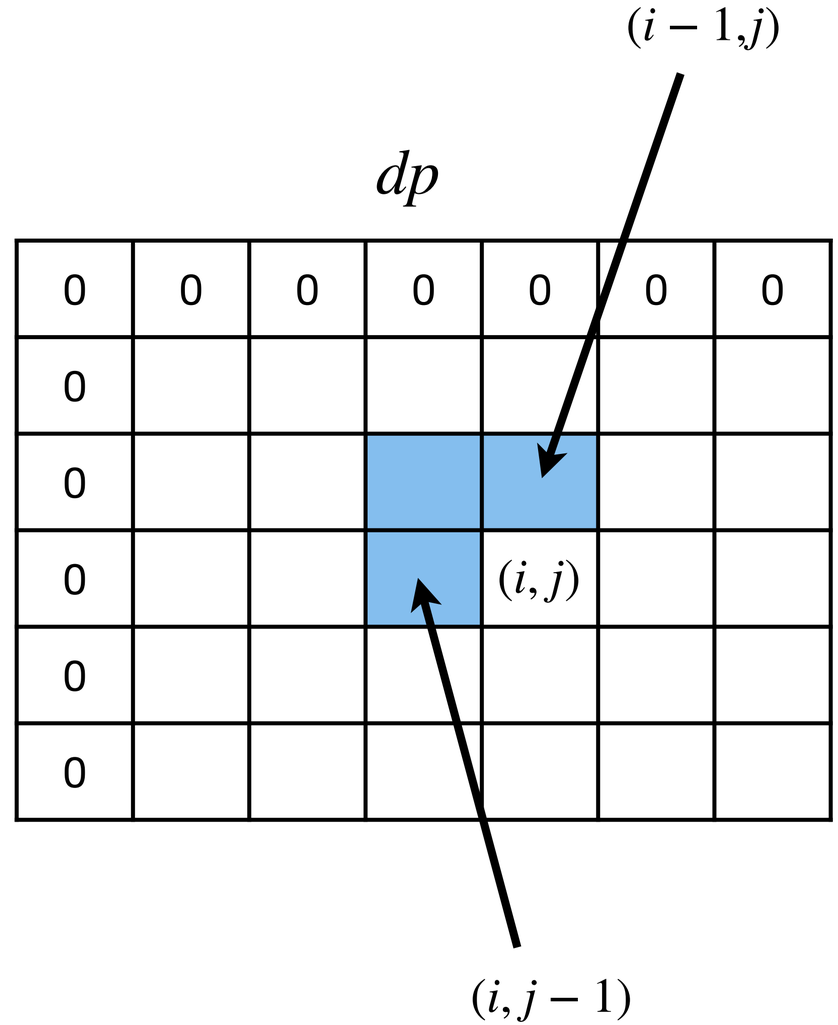

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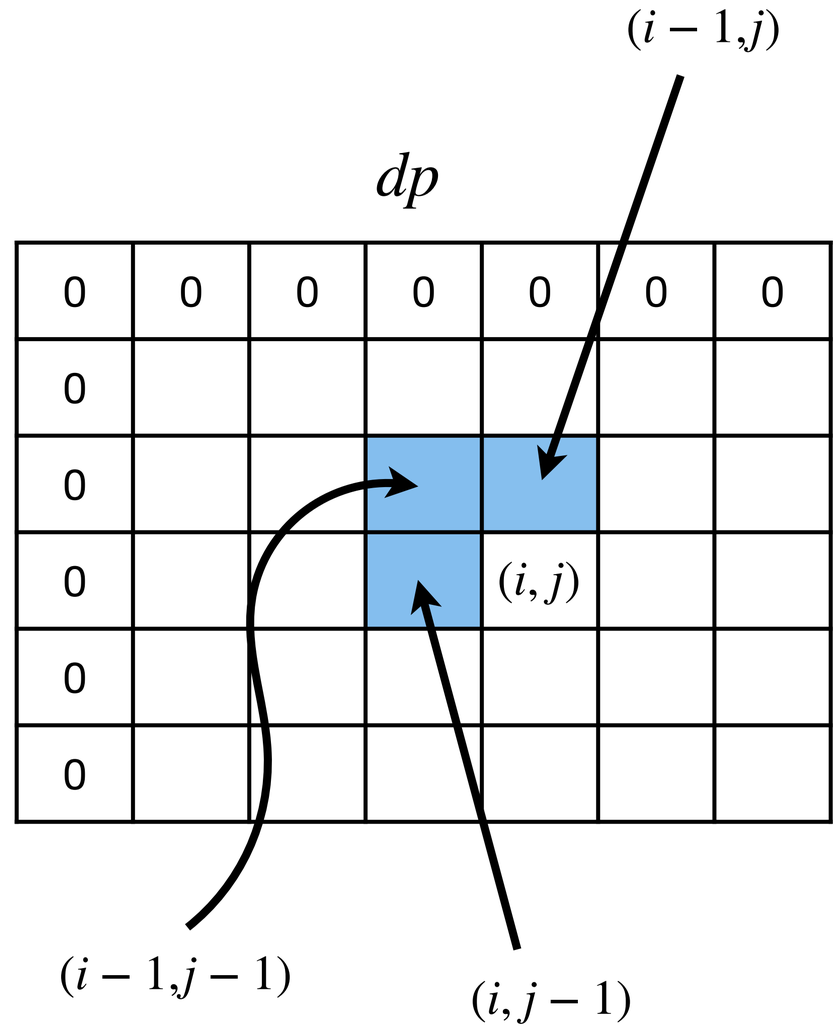

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Time Complexity: $O(nm)$

Bottom-Up vs Top-Down DP

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- Top-down solves only those subproblems which are necessary.

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- Bottom-up solves all.
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- Otherwise, then top down is better.